

Reduced Graphene Oxide/LiI Composite Lithium Ion Battery Cathodes

Sanghyeon Kim,^{†,‡,§} Sung-Kon Kim,^{†,‡,§,||} Pengcheng Sun,^{†,‡,§} Nuri Oh,^{†,§} and Paul V. Braun^{*,†,‡,§,||}

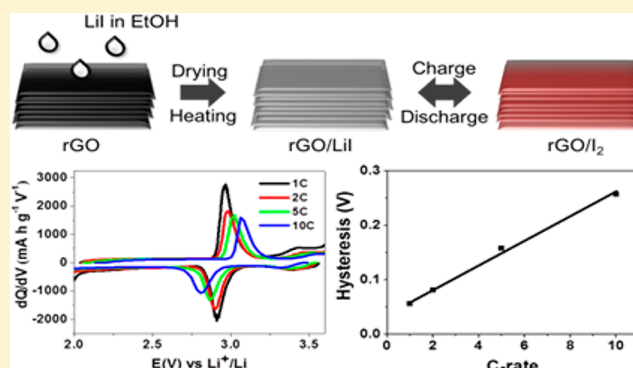
[†]Department of Materials Science and Engineering, [‡]Frederick Seitz Materials Research Laboratory, and [§]Beckman Institute for Advanced Science and Technology, University of Illinois at Urbana–Champaign, Urbana, Illinois 61801, United States

^{||}School of Chemical Engineering, Chonbuk National University, 567 Baekje-daero, Deokjin-gu, Jeonju-si, Jeollabuk-do 54896, Republic of Korea

Supporting Information

ABSTRACT: Li-iodine chemistry is of interest for electrochemical energy storage because it has been shown to provide both high power and high energy density. However, Li-iodine batteries are typically formed using Li metal and elemental iodine, which presents safety and fabrication challenges (e.g., the high vapor pressure of iodine). These disadvantages could be circumvented by using LiI as a starting cathode. Here, we present fabrication of a reduced graphene oxide (rGO)/LiI composite cathode, enabling for the first time the use of LiI as the Li-ion battery cathode. LiI was coated on rGO by infiltration of an ethanolic solution of LiI into a compressed rGO aerogel followed by drying. The free-standing rGO/LiI electrodes show stable long-term cycling and good rate performance with high specific capacity (200 mAh g^{−1} at 0.5 C after 100 cycles) and small hysteresis (0.056 V at 1 C). Shuttling was suppressed significantly. We speculate the improved electrochemical performance is due to strong interactions between the active materials and rGO, and the reduced ion and electron transport distances provided by the three-dimensional structured cathode.

KEYWORDS: Lithium iodine battery, lithium iodide, reduced graphene oxide aerogel, cathode, hysteresis, shuttling



Demand for high energy and power density secondary batteries for a range of technologies, including next-generation hybrid and all-electric vehicles,^{1–3} has led to considerable research in batteries, which provide both high power and charge acceptance, without a significant reduction in energy. The Li-iodine system ($I_2 + 2Li \leftrightarrow 2LiI$) has been considered promising because of fast electrochemical conversion of the iodine/triiodide redox pair and high reaction potential with lithium.^{4–6} However, to date, there have been only a few reports on rechargeable Li-iodine cells^{7–11} because of several issues including the following: iodine is highly soluble in aprotic electrolytes, leading to self-discharge due to shuttling;⁷ Li-iodine cells generally use Li metal as an anode, however, Li metal can form dendrites during cycling that lead to shorting;¹² iodine is volatile even at room temperature, leading to cell fabrication challenges;⁸ and iodine based electrodes suffer from poor electrical conductivity.

Here, we present the fabrication of a reduced graphene oxide (rGO)/LiI composite cathode with a gravimetric specific capacity of 200 mAh g^{−1} after 100 cycles at 0.5 C and stable cycling performance at both low (0.5 C) and high (10 C) current densities. Promisingly, the rGO/LiI electrodes show good rate performance and a small hysteresis even at high C-rates (0.257 V at 10 C), which addresses some of the problems

with the Li-iodine system, providing hope that the Li-iodine chemistry may be viable for energy storage. While LiI has been studied as an electrolyte additive or as a component of a solid electrolyte,^{13–16} to the best of our knowledge, this is the first report of LiI being used as a secondary lithium ion battery electrode material. We chose the rGO as the host for LiI because of its high electrical conductivity, porous nature, and good adsorptivity for both LiI and I₂.^{12,17,18} Specifically, the oxygen-containing functional groups in rGO such as carbonyl and alkoxy groups, are thought to provide strong anchoring points to other elements, resulting in, for example, good lithium affinity.^{19–21} For example, rGO was found to serve as a shuttle inhibitor in lithium–sulfur batteries.¹⁹ Additionally, rGO can suppress the dissolution of active materials species during cycling and has also been found to facilitate rapid electron transfer.^{12,17,18} Importantly, the use of LiI as the active cathode material rather than elemental iodine enables use of lithium-free anodes such as graphite, silicon, and tin. The high melting (469 °C) and boiling points (1171 °C) of LiI provide better thermostability compared to iodine (melting point, 113.7 °C;

Received: August 1, 2017

Revised: October 2, 2017

Published: October 20, 2017



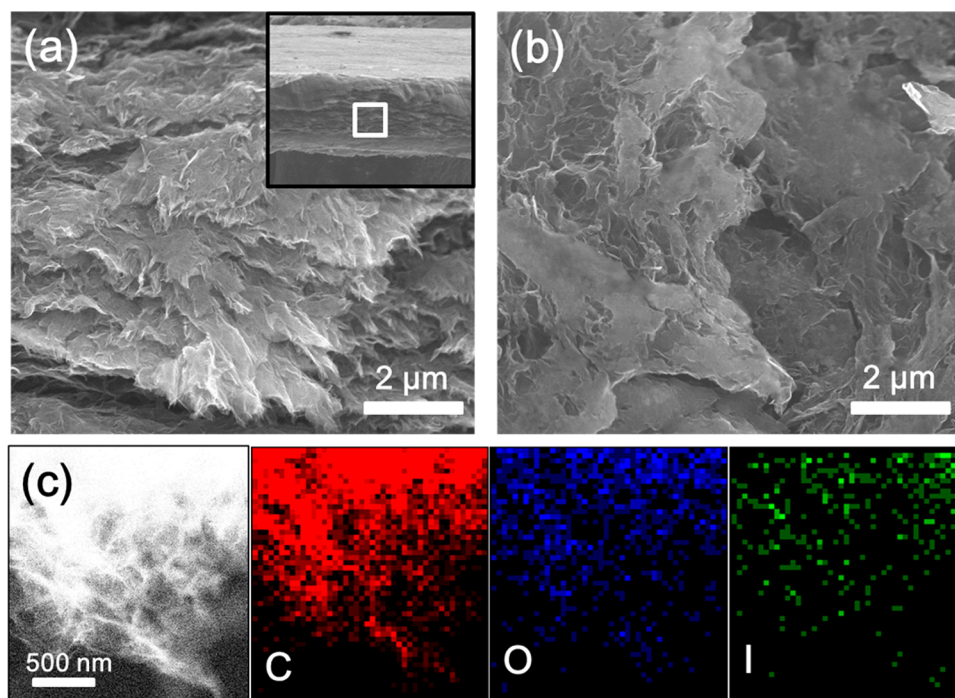


Figure 1. Cross-sectional SEM images of (a) bare rGO and (b) a rGO/LiI electrode. The inset in panel a is a low magnification cross-sectional SEM image of the bare rGO indicating where the high magnification image was taken (white box). (c) HAADF-STEM image of the rGO/LiI electrode and corresponding EDS mapping for C, O, and I.

boiling point, 184 °C), enabling heating during cell fabrication as necessary.

The LiI active materials were loaded into a compressed rGO aerogel from an ethanolic LiI solution. As ethanol evaporates, LiI precipitates in the rGO. The resulting rGO/LiI composite was directly used as a free-standing electrode without a current collector or additional binder. Cross-sectional scanning electron microscopy (SEM) images of bare rGO and rGO/LiI composites are shown in Figure 1a,b. Cross-sectional SEM images were taken at the center of rGO film to ensure that LiI is uniformly distributed throughout rGO film. Figure 1a shows rGO is made up of stacks of wrinkled graphene sheets and that the sheets are porous, which are beneficial for both solution infiltration and ion transport. Figure 1b shows an SEM image of the rGO/LiI electrode. No large LiI particles were observed, perhaps because the LiI coats the rGO rather uniformly. To further confirm uniform coating of rGO with LiI, energy dispersive X-ray spectroscopy (EDS) mapping using aberration-corrected scanning transmission electron microscopy was conducted (Figure 1c). The EDS mapping indicates that C, O, and I are uniformly distributed in the rGO/LiI composite. The relatively low intensity of I might be due to the loss of I during STEM sample preparation.

X-ray diffraction (XRD) spectra of the rGO and rGO/LiI electrodes are shown in Figure 2a. rGO exhibited a broad 002 peak at $2\theta = 23.5^\circ$, coming from the graphene layers.²² The rGO/LiI composite showed no diffraction peak other than two broad peaks coming from the Kapton tape (necessary to protect the LiI from oxidation), indicating the precipitated LiI is either nanocrystalline or amorphous. If a broad rGO peak is present, it may be hidden by the Kapton peaks. X-ray photoelectron spectroscopy (XPS) was carried out to confirm the presence of LiI on rGO (Figure 2b,c) and investigate the interaction between LiI and rGO (Figure S2). The Li 1s peak at about 55.9 eV corresponds to Li^+ .²³ The two peaks observed at

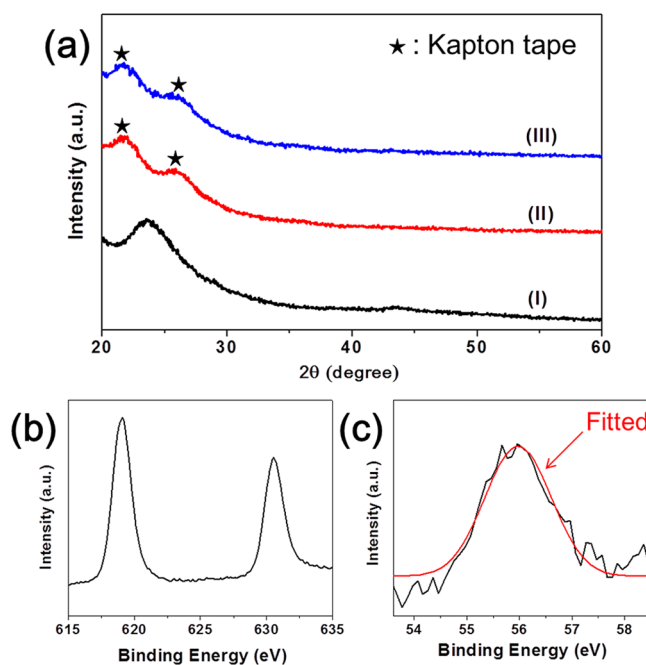
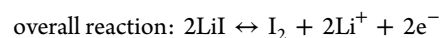


Figure 2. (a) XRD of (I) rGO without kapton tape, (II) rGO, and (III) rGO/LiI with kapton tape. XPS spectra of (b) I 3d and (c) Li 1s from rGO/LiI.

619.1 and 630.5 eV in the I 3d spectra matches the characteristic peaks of I^- .²⁴ No new peaks or significant peak shift were observed in the C 1s XPS spectra of rGO/LiI relative to the starting rGO, an indication that the interaction between LiI and C is mostly physical. Interestingly, the O 1s XPS peak shifted about 1 eV toward higher energy (to about 533 eV) after LiI infilling (Figure S3). This is perhaps because electrons in oxygen are attracted to Li^+ , leading to a higher binding

energy.²⁵ Li_2O 's O 1s peak is at 528.6 eV,^{23,26} so while Li_2O can form in the process of XPS measurement, no Li_2O formation is observed in this study. The interaction between residual oxygen in the rGO and Li in the LiI perhaps improves adsorption of LiI on rGO during cycling. The oxygen content in rGO is found to be about 10% by EDS.

Cyclic voltammetry (CV) was performed on the rGO/LiI composite to investigate its electrochemical properties (Figure 3a). In the anodic scan, LiI is oxidized to LiI_3 at 2.88 V, and LiI_3



To study the kinetics of the LiI electrodes, CV was performed at scan speeds from 0.5 to 2 mV s^{-1} (Figure 3b). The second anodic peaks are missing in Figure 3b at high scan rates. This is because the peak potential moves toward higher potentials at high scan rates unless the reaction is Nernstian in which charge transfer resistance is zero.²⁷ It is known that peak current has the following relationship with scan rate: $I_p = av^b$ where a is a constant and v is the scan rate. b ranges from 1/2 to 1, depending on whether the reaction is diffusion controlled or a capacitive process.^{28,29} Figure 3c shows a logarithmic relationship between peak currents and scan rate. b was determined to be 0.6 and 0.54 for the anodic and cathodic reactions, respectively, indicating the LiI_3/LiI redox pair is close to diffusion-controlled.

The formation of I_2 , final product was confirmed by an ex situ XPS measurement on a charged LiI/rGO electrode (Figure S4). An $\text{I } 3d_{5/2}$ peak was observed at 619.9 eV, which matches the characteristic peak of elemental iodine.^{30,31} The $\text{I } 3d_{5/2}$ XPS peak shifted about 0.8 eV toward higher energy after charge process (Figures 2b and S4), supporting the hypothesis that LiI was converted to I_2 upon charging.

The electrochemical properties of the rGO/LiI electrodes were evaluated by galvanostatic discharge/charge tests at room temperature. Figure 4a shows the performance of the rGO/LiI electrodes at 0.5 C over 100 cycles. Prior to extended cycling, the electrodes were precycled three times at 1 C to enhance the electrode stability. The initial specific capacity of the rGO/LiI electrodes is 270 mA h g^{-1} , which is higher than theoretical capacity of LiI (200 mA h g^{-1}). This extra capacity comes from the rGO. The bare rGO capacity is about 30 mA h g^{-1} over the voltage range of 2–3.6 V (Figure S5), and the weight fraction of LiI in the composite is 30–40%. Since specific capacity is calculated based on the weight of LiI only, an increase in specific capacity of 70 mA h g^{-1} from the rGO is reasonable. Charge capacity is always higher than discharge capacity because of shuttling. Figure 4b shows two pairs of plateaus, which are consistent with the CV profiles. The shuttling was also observed in discharge/charge curves through longer reaction contributes more to shuttling, cycling tests were conducted at cutoff voltages of 2.0–3.3 and 2.0–3.6 V (Figure S6). When the electrode was cycled in the voltage range of 2.0–3.3 V, little shuttling was observed, indicating shuttling during the LiI_3/I_2 reaction is more prominent than for LiI/LiI_3 . This perhaps because I_2 is more soluble than LiI_3 in the DOL/DME electrolyte and as they will have reduced Coulombic attraction with O in the rGO.

While there is shuttling at high voltages, other than this issue, the electrodes showed rather good cycling stability. The average capacity decay rate per cycle over 100 cycles is 0.28%, and as low as 0.13% over cycles 50 to 100. We attribute the high capacity and good cycling stability to the rGO, which both provides strong absorption of the active materials and a conductive pathway for electrons. To confirm the effect of rGO, a control experiment was performed using a carbon nanofiber (CNF) film rather than rGO. CNF/LiI electrodes were made using the same fabrication process, and cycling was performed at 0.5 C in the voltage range of 2.0–3.6 V (Figure S7). The CNF/LiI electrodes exhibited low specific capacity ($\sim 40 \text{ mA h g}^{-1}$). Furthermore, the color of the electrolyte turned yellow after cycling, in contrast to the electrolyte for rGO/LiI, which

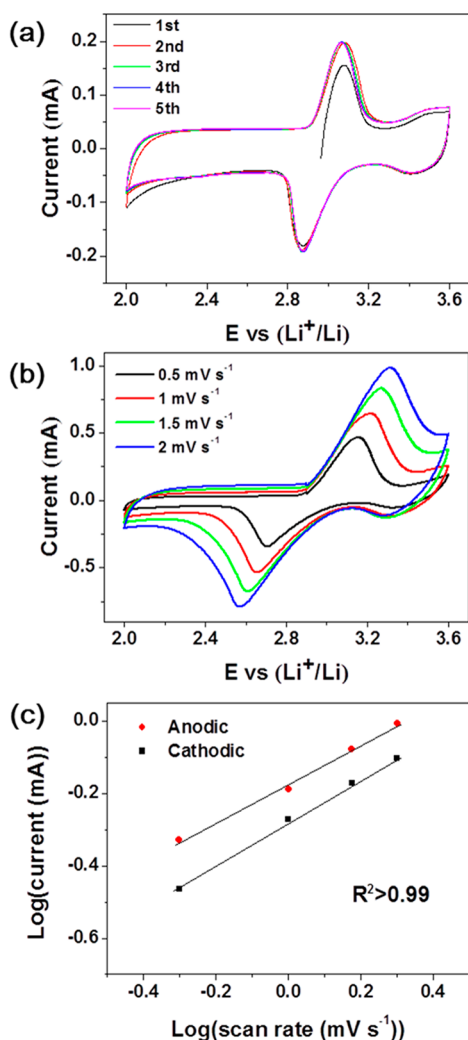
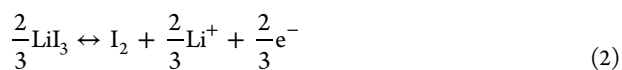
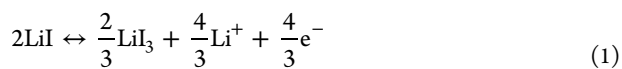


Figure 3. (a) Cyclic voltammetry curves of the rGO/LiI electrode at a scan rate of 0.5 mV s^{-1} and (b) scan rates of 0.5, 1, 1.5, and 2 mV s^{-1} . (c) Plot of $\log(\text{peak current})$ vs $\log(\text{scan rate})$ obtained from panel b.

is further oxidized to I_2 at 3.5 V.^{7–11} The reason why the current at the second anodic peak at 3.5 V looks saturated is because of shuttling. The reverse reaction occurs in the cathodic scan. I_2 is reduced to LiI_3 at 3.4 V, and LiI_3 is further reduced to LiI at 3.07 V. All oxidation and reduction peaks overlap well after the first cycle, indicating stable and reversible electrochemical reactions. The battery reactions can be described as follows:



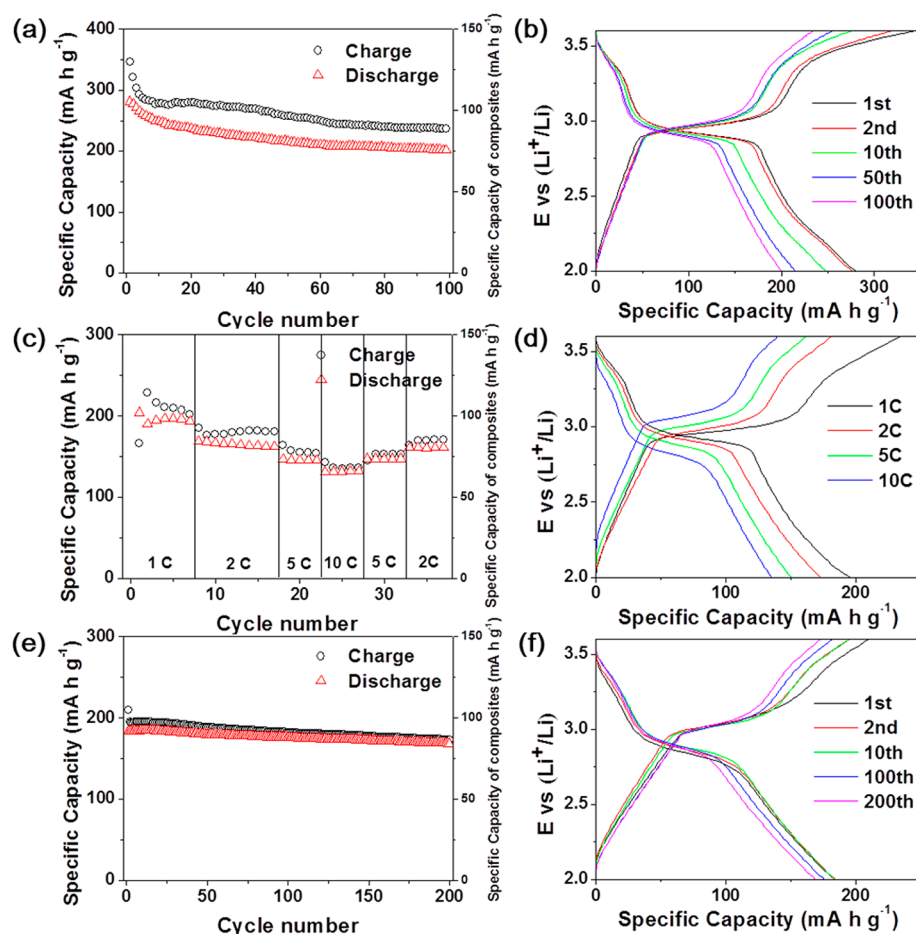


Figure 4. (a) Cycling performance of rGO/LiI cathode at 0.5 C. (b) First, second, 10th, 50th, and 100th cycle discharge–charge curves of rGO/LiI at 0.5 C. (c) Rate performance of LiI at 1 to 10 C. (d) Second cycle rGO/LiI discharge/charge curves at different C-rates. (e) Cycling performance of rGO/LiI at 10 C. (f) First, second, 10th, 100th, and 200th cycle rGO/LiI discharge/charge curves at 0.5 C.

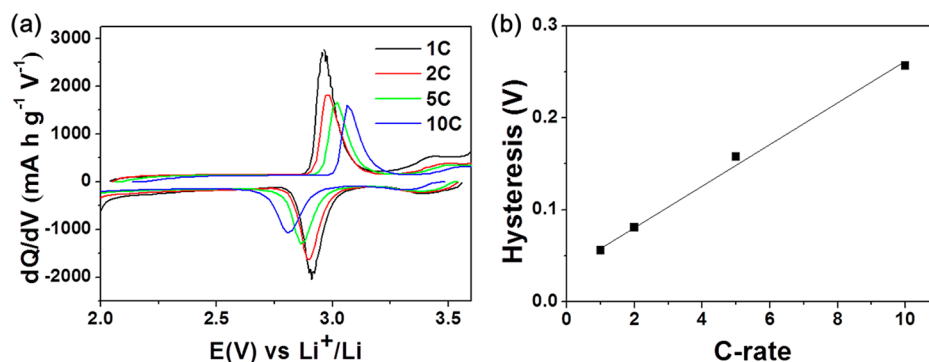


Figure 5. (a) Differential capacity plot of the second cycle at 1, 2, 5, and 10 C. (b) Plot of C-rate vs hysteresis (V) based on panel a.

remained clear after cycling. The electrolyte color change is due to LiI_3 and I_2 species, which dissolve during cycling.⁸ This result agrees with previous research, which has shown that heteroatoms incorporated in graphene can offer strong binding sites for polar ionic compounds, resulting in significantly improved electrochemical performance similar to what we observe.^{12,17,25,32}

The rate performance of the rGO/LiI electrodes was evaluated by cycling at different C-rates. The 2 C capacity was 88% of the 1 C capacity, the 5 C capacity was 77% of the 1 C capacity, and even at 10 C, the electrode delivered 69% of the 1 C capacity (Figure 4d). Moreover, when the rate returns back

to 1 C, most capacity was recovered. The reason for the high-rate performance will be discussed later.

Figure 4e shows the cycling performance of rGO/LiI electrode at 5 C after precycling at 1 C for three cycles. After 200 cycles, the discharge capacity was 168 mA h g^{-1} . Interestingly, an electrode cycled at 10 C showed even better cycling stability, and at this C-rate, no significant shuttling was observed. The shuttling during the LiI_3/I_2 redox reaction was restricted at this C-rate probably because there was not sufficient time for significant amounts of active material to dissolve, travel to, and react with lithium. The difference in the specific capacities (36 mA h g^{-1}) between the samples

presented in Figure 4c,e is probably due to variations in the rGO/LiI weight ratio (60:40 for Figure 4c, 67:33 for Figure 4e) and the fact that the rGO has a non-negligible capacity, and the capacity loss of the sample presented in Figure 4c is due to it being cycled 22 times at lower C-rates before testing at 10 C, while the sample presented in Figure 4e had only been cycled three times at lower C-rates before testing at 10 C.

Importantly, there are few changes in the voltage plateaus (Figure 4b) with increasing C-rate. The voltage hysteresis at 1, 2, 5, and 10 C was determined from dQ/dV curves (Figure 5a) obtained by differentiating the charge/discharge voltage curves of the second cycle at each C-rate. Two pairs of redox peaks observed in cyclic voltammetry were also seen in the dQ/dV curves. The hysteresis was taken as the difference between the LiI_3/LiI redox peaks. Figure 5b presents the calculated hysteresis, 0.056, 0.081, 0.158, and 0.257 V, at 1, 2, 5, and 10 C, respectively. These values are the smallest we are aware of for alloying or conversion reactions (a similar trend is observed for the I_2/LiI_3 redox couple, where the hysteresis is 0.047, 0.117, 0.108, and 0.277 V at 1, 2, 5, and 10 C, respectively). Hysteresis is the sum of polarization in charge and discharge processes. The primary contributions to polarization are (1) activation polarization related to charge transfer reactions, (2) ohmic polarization coming from internal resistances of cell components and the contact resistance between cell components, and (3) concentration polarization coming from limitations in diffusion.³³ Figure 3c shows the reaction kinetics of the LiI_3/LiI redox pair is mostly limited by diffusion rather than charge transfer indicating concentration polarization is more important than activation polarization. The ohmic polarization has a linear relationship with current ($\eta = IR$), while concentration polarization is logarithmic in terms of current ($\eta = c \log\left(\frac{I_i}{I_i - I}\right)$ where c is constant). Since hysteresis was linear with current (C-rate), it appears the (small) hysteresis is mostly dominated by ohmic polarization. When concentration polarization is dominant, hysteresis will not be linear with current, and the point where the hysteresis becomes nonlinear with current is dependent on the limiting current (I_l), the current where the Li concentration gradient becomes steep within the active materials. As shown in the above equation, the higher the limiting current, the smaller the concentration polarization. The limiting current is defined as follows:

$$I_l = \frac{nFAD_0C_0^*}{\delta} \quad (D_0, \text{diffusion coefficient; } C_0^*, \text{concentration of Li at the electrode surface; } \delta, \text{diffusion length}).$$

The limiting current of the rGO/LiI electrode is expected to be high because of short solid-state diffusion lengths (LiI particle thickness) and high Li diffusion coefficient in LiI.^{34,35} The fact that the activation polarization and concentration polarization is not changing significantly with current density is a sign that charge transfer and mass transfer effects are not limiting the rate performance, resulting in the high-rate performance.

To examine the cause of capacity fade of the rGO/LiI electrodes, impedance measurements were conducted after 10, 20, and 30 cycles (Figure 6). The resulting three Nyquist plots consist of two depressed semicircles in the high and medium frequency regions and an inclined line indicating solid-state Li diffusion in the low frequency regions. The charge transfer resistance is shown by the diameters of the two semicircles. The presence of two semicircles in the Nyquist plot is presumably due to dissolved redox species.³⁶ Marcus theory suggests charge transfer reaction rates are affected by environment.²⁷ Charge

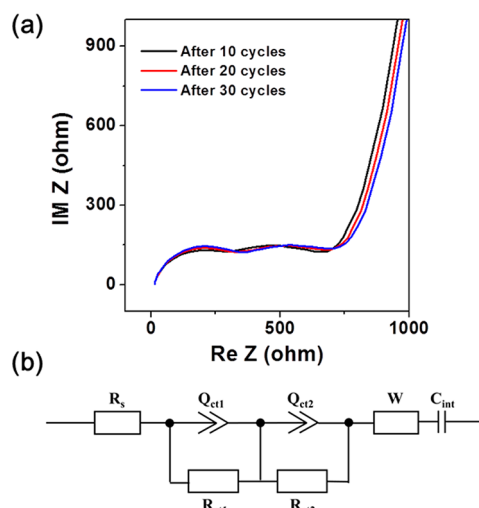


Figure 6. (a) Electrochemical impedance spectroscopy of rGO/LiI electrodes after 10, 20, and 30 cycles. (b) Equivalent circuit model used for impedance data fitting. (R_s , solution resistance; R_{ct1} and R_{ct2} , charge transfer resistance; Q_{ct1} Q_{ct2} , constant phase element; W , Warburg impedance; C_{int} , intercalation capacitance).

transfer reactions of dissolved redox species should be fast because structural reorganization in the electrolyte is easier.^{27,36} Therefore, the high frequency semicircle may represent electrochemical redox reactions of dissolved species and the medium frequency semicircle related to redox reactions on the solid electrode. One concern is that LiNO_3 has been reported to react with lithium to form a protective surface film.³⁷ To ensure this film has no impact on the impedance measurements, they were repeated immediately after replacing the Li metal anode with a new one. The same two semicircles were still observed. Considering the LiNO_3 -derived film is not formed before cycling,³⁸ this result validates the two semicircles are not related to the Li anode.

All Nyquist plots showed significantly overlapped shapes and similar charge transfer resistances (Figure 6a and Table S2), indicating the electrode reaction kinetics remained largely unchanged during cycling. For further insight into the capacity fade, SEM analysis was conducted on the Li metal anode after cycling. After cycling 30 times, the Li metal anode was removed from the electrolyte and dried in a glovebox without washing. Figure S8a shows the SEM image of the anode after cycling. Small particles were observed on the anode. In lithium–sulfur batteries, the dissolved polysulfides can migrate throughout the separator by shuttling to form insoluble sulfides (Li_2S_2 , Li_2S) onto Li surface. The gradual growth of this inactive layer contributes to the capacity decay by consuming active materials.^{39–41} A similar phenomenon was observed here. EDS (Figure S8b) indicates these particles contain O, S, and I, indicating iodine species were consumed by shuttling and forming compounds on the Li surface, providing the possibility the capacity decay with cycling is due to loss of active materials. As a control experiment, an uncycled Li metal anode was treated similarly, and no particles were observed on the anode, indicating the particles form during cycling. The LiI morphology after cycling (10 cycles, 0.5 C) is not clear, similar to the previously reported unclear morphology of a rGO/ Li_2S electrode after cycling (Figure S9). The EDS spectrum and iodine mapping indicate LiI is present on the rGO (Figure S9b and S9c). Electrolyte (F, O, and S peaks) were also observed in

the EDS spectrum as the electrode was not washed before EDS analysis because of high solubility of LiI in organic electrolyte.

In conclusion, rGO/LiI electrodes prepared by a simple solution infiltration, evaporation, and precipitation method were demonstrated as a lithium ion battery cathode for the first time. The rGO/LiI cathode shows a reasonable specific capacity and stable cycling behavior at both low and high current densities probably because rGO provides an efficient electron pathway and suppresses the dissolution of active materials during cycling. The rGO/LiI electrode exhibits good rate performance and small hysteresis at high C-rate, which can be attributed to its fast reaction kinetics resulting from the short electron and ion diffusion lengths in this system. While we have yet to do the experiments, we suspect the performance of rGO/LiI electrodes can be increased further by engineering the rGO oxygen content or by doping another element into the rGO and by electrolyte optimization. This work demonstrates the possibility of using LiI as a battery cathode and may inspire further investigations on LiI-based cathodes.

Experimental Section. *Synthesis of rGO Aerogel.* GO was synthesized by a modified Hummer's method using expanded graphite.⁴² The rGO aerogel was synthesized as reported.⁴³ GO (4 mg/mL) was dispersed into an aqueous solution, which has the weight ratio of GO/hypophosphorous acid (H_3PO_2)/ I_2 of 1:100:10. The final volume was 40 mL. The solution was sonicated for 5 min and then put into a water-bath maintained at 90 °C. A black gel-like cylinder was removed after 12 h and washed with alcohol and water in a Soxhlet extractor for 12 h. The wet gels were freeze-dried for 2 days to obtain the aerogel.

Synthesis of rGO/Li Composite. A 0.1 M LiI solution was prepared by dissolving lithium iodide (LiI, Sigma-Aldrich) in anhydrous ethanol with stirring overnight in an argon-filled glovebox. The rGO aerogels were pressed and cut into about 40 μm -thick pieces and dried at 100 °C under vacuum. The resulting rGO electrodes were then transferred into the glovebox. Since LiI is air and moisture sensitive, the remaining steps were all done in a glovebox. LiI solution was dropped onto the rGO electrodes using a pipet and the ethanol fully evaporated. The rGO electrodes were then turned over, and the LiI-infilling step repeated. The whole process was repeated two additional times. The rGO/LiI composite materials were then heated at 150 °C for 1 h. The areal capacity was around 0.4 mA h cm^{-2} . The areal capacity can be further increased by pressing multiple electrodes together (Figure S11).

Characterization. Electrode morphologies were confirmed using a Hitachi S-4700 SEM. EDS was conducted using a Hitachi S-4700 SEM equipped with an Oxford INCA energy-dispersive X-ray analyzer. XPS analysis were done with a Kratos Axis Ultra XPS system with a monochromatic Al K α (1486.6 eV) source. Binding energies were calibrated with respect to the C 1s peak (284.4 eV). To minimize air exposure time, samples for SEM and XPS measurement were prepared in the glovebox. The sample was attached on the sample holder, which was put in a bottle filled with Ar gas and tightly sealed. The sample holder was taken out of the bottle just before transferring to the vacuum chamber. The vacuum chamber was immediately pumped down. The air exposure time was less than 1 s. The crystal structures were checked by a Philips X'pert MRD XRD (Materials Research Diffractometer) using Cu K α radiation (λ = 0.15418 nm). The rGO/LiI sample was sealed in the glovebox using Kapton tape to prevent oxidation of LiI. STEM-EDS was performed using JEOL 2200FS for elemental mapping.

Electrochemical Measurements. Lithium bis-(trifluoromethane sulfonyl) imide (LiTFSI, 1.0 M) in 1,3-dioxolane (DOL)/1,2-dimethoxyethane (DME) (1:1 by vol) containing 1 wt % LiNO_3 was used as electrolyte and lithium foil was used as the counter-electrode for two-electrode cells. All cells were assembled in a glovebox. Galvanostatic discharge/charge tests and cyclic voltammetry were conducted using a VMP3, Bio-Logic potentiostat and swagelok-type cells over a voltage range of 2.0 to 3.6 V or 2.0 to 3.3 V vs Li/Li⁺. Current densities and specific capacities were calculated based on LiI. An AC amplitude of 6 mV was applied over the frequency range of 100 kHz to 10 mHz for impedance measurements.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.nanolett.7b03290.

Additional information and figures (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: pbraun@illinois.edu.

ORCID

Nuri Oh: 0000-0001-9145-8911

Paul V. Braun: 0000-0003-4079-8160

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the U.S. Department of Energy, Office of Basic Energy Sciences, Division of Materials Sciences and Engineering under Award # DE-FG02-07ER46471, through the Frederick Seitz Materials Research Laboratory at the University of Illinois at Urbana-Champaign. S.K. would like to show appreciation to the Kwanjeong Educational Foundation for a scholarship.

■ REFERENCES

- (1) Armand, M.; Tarascon, J.-M. *Nature* **2008**, *451*, 652–657.
- (2) Lee, S. W.; Yabuuchi, N.; Gallant, B. M.; Chen, S.; Kim, B.-S.; Hammond, P. T.; Shao-Horn, Y. *Nat. Nanotechnol.* **2010**, *5*, 531–537.
- (3) Etacheri, V.; Marom, R.; Elazari, R.; Salitra, G.; Aurbach, D. *Energy Environ. Sci.* **2011**, *4*, 3243–3262.
- (4) Behl, W. K.; Chin, D. T. *J. Electrochem. Soc.* **1988**, *135*, 16–21.
- (5) Boschloo, G.; Hagfeldt, A. *Acc. Chem. Res.* **2009**, *42*, 1819–1826.
- (6) Dong, X.; Chen, L.; Liu, J.; Haller, S.; Wang, Y.; Xia, Y. *Sci. Adv.* **2016**, *2*, e1501038.
- (7) Wang, Y.; Sun, Q.; Zhao, Q.; Cao, J.; Ye, S. *Energy Environ. Sci.* **2011**, *4*, 3947–3950.
- (8) Zhao, Q.; Lu, Y.; Zhu, Z.; Tao, Z.; Chen, J. *Nano Lett.* **2015**, *15*, 5982–5987.
- (9) Li, K.; Lin, B.; Li, Q.; Wang, H.; Zhang, S.; Deng, C. *ACS Appl. Mater. Interfaces* **2017**, *9*, 20508–20518.
- (10) Wang, H.; Zhang, G.; Ke, L.; Liu, B.; Zhang, S.; Deng, C. *Nanoscale* **2017**, *9*, 9365–9375.
- (11) Zhang, Q.; Wu, Z.; Liu, F.; Liu, S.; Liu, J.; Wang, Y.; Yan, T. *J. Mater. Chem. A* **2017**, *5*, 15235–15242.
- (12) Zhou, G.; Paek, E.; Hwang, G. S.; Manthiram, A. *Adv. Energy Mater.* **2016**, *6*, 1501355.
- (13) Liu, F.-C.; Liu, W.-M.; Zhan, M.-H.; Fu, Z.-W.; Li, H. *Energy Environ. Sci.* **2011**, *4*, 1261–1264.

- (14) Park, K. H.; Oh, D. Y.; Choi, Y. E.; Nam, Y. J.; Han, L.; Kim, J. Y.; Xin, H.; Lin, F.; Oh, S. M.; Jung, Y. S. *Adv. Mater.* **2016**, *28*, 1874–1883.
- (15) Park, S. H.; Kim, H. J.; Jeon, J.; Choi, Y.; Cho, J. J.; Lee, H. *ChemElectroChem* **2016**, *3*, 1915–1921.
- (16) Wu, F.; Lee, J. T.; Nitta, N.; Kim, H.; Borodin, O.; Yushin, G. *Adv. Mater.* **2015**, *27*, 101–108.
- (17) Ji, L.; Rao, M.; Zheng, H.; Zhang, L.; Li, Y.; Duan, W.; Guo, J.; Cairns, E. J.; Zhang, Y. *J. Am. Chem. Soc.* **2011**, *133*, 18522–18525.
- (18) Wang, C.; Wang, X.; Yang, Y.; Kushima, A.; Chen, J.; Huang, Y.; Li, J. *Nano Lett.* **2015**, *15*, 1796–1802.
- (19) Wang, X.; Wang, Z.; Chen, L. *J. Power Sources* **2013**, *242*, 65–69.
- (20) Hu, X.; Li, Z.; Zhao, Y.; Sun, J.; Zhao, Q.; Wang, J.; Tao, Z.; Chen, J. *Sci. Adv.* **2017**, *3*, e1602396.
- (21) Lin, D.; Liu, Y.; Liang, Z.; Lee, H.-W.; Sun, J.; Wang, H.; Yan, K.; Xie, J.; Cui, Y. *Nat. Nanotechnol.* **2016**, *11*, 626–632.
- (22) Stobinski, L.; Lesiak, B.; Malolepszy, A.; Mazurkiewicz, M.; Mierzwa, B.; Zemek, J.; Jiricek, P.; Bielloshapka, I. *J. Electron Spectrosc. Relat. Phenom.* **2014**, *195*, 145–154.
- (23) Tanaka, S.; Taniguchi, M.; Tanigawa, H. *J. Nucl. Mater.* **2000**, *283*, 1405–1408.
- (24) Strydom, C.; van Staden, J.; Strydom, H. *Electroanalysis* **1991**, *3*, 197–201.
- (25) Yang, C.-P.; Yin, Y.-X.; Ye, H.; Jiang, K.-C.; Zhang, J.; Guo, Y.-G. *ACS Appl. Mater. Interfaces* **2014**, *6*, 8789–8795.
- (26) Yao, K. P.; Kwabi, D. G.; Quinlan, R. A.; Mansour, A. N.; Grimaud, A.; Lee, Y.-L.; Lu, Y.-C.; Shao-Horn, Y. *J. Electrochem. Soc.* **2013**, *160*, A824–A831.
- (27) Bard, A. J.; Faulkner, L. R.; Leddy, J.; Zoski, C. G. *Electrochemical Methods: Fundamentals and Applications*, 2nd ed.; Wiley, 2001.
- (28) Augustyn, V.; Come, J.; Lowe, M. A.; Kim, J. W.; Taberna, P.-L.; Tolbert, S. H.; Abruña, H. D.; Simon, P.; Dunn, B. *Nat. Mater.* **2013**, *12*, 518–522.
- (29) Brezesinski, T.; Wang, J.; Tolbert, S. H.; Dunn, B. *Nat. Mater.* **2010**, *9*, 146–151.
- (30) Gong, D.; Wang, B.; Zhu, J.; Podila, R.; Rao, A. M.; Yu, X.; Xu, Z.; Lu, B. *Adv. Energy Mater.* **2017**, *7*, 1601885.
- (31) Tian, H.; Gao, T.; Li, X.; Wang, X.; Luo, C.; Fan, X.; Yang, C.; Suo, L.; Ma, Z.; Han, W. *Nat. Commun.* **2017**, *8*, 14083.
- (32) Peng, H. J.; Hou, T. Z.; Zhang, Q.; Huang, J. Q.; Cheng, X. B.; Guo, M. Q.; Yuan, Z.; He, L. Y.; Wei, F. *Adv. Mater. Interfaces* **2014**, *1*, 1400227.
- (33) Winter, M.; Brodd, R. J. *Chem. Rev.* **2004**, *104*, 4245–4269.
- (34) Liu, F.-C.; Shadik, Z.; Wang, X.-F.; Shi, S.-Q.; Zhou, Y.-N.; Chen, G.-Y.; Yang, X.-Q.; Weng, L.-H.; Zhao, J.-T.; Fu, Z.-W. *Inorg. Chem.* **2016**, *55*, 6504–6510.
- (35) Cao, Z.; Liu, Z.; Sun, J.; Huang, F.; Yang, J.; Wang, Y. *Solid State Ionics* **2008**, *179*, 1776–1778.
- (36) Yang, Y.; Zheng, G.; Misra, S.; Nelson, J.; Toney, M. F.; Cui, Y. *J. Am. Chem. Soc.* **2012**, *134*, 15387–15394.
- (37) Aurbach, D.; Pollak, E.; Elazari, R.; Salitra, G.; Kelley, C. S.; Affinito, J. *J. Electrochem. Soc.* **2009**, *156*, A694–A702.
- (38) Liang, X.; Wen, Z.; Liu, Y.; Wu, M.; Jin, J.; Zhang, H.; Wu, X. *J. Power Sources* **2011**, *196*, 9839–9843.
- (39) Evers, S.; Nazar, L. F. *Acc. Chem. Res.* **2012**, *46*, 1135–1143.
- (40) Ji, X.; Nazar, L. F. *J. Mater. Chem.* **2010**, *20*, 9821–9826.
- (41) Manthiram, A.; Fu, Y.; Su, Y.-S. *Acc. Chem. Res.* **2012**, *46*, 1125–1134.
- (42) Pham, V. H.; Cuong, T. V.; Nguyen-Phan, T.-D.; Pham, H. D.; Kim, E. J.; Hur, S. H.; Shin, E. W.; Kim, S.; Chung, J. S. *Chem. Commun.* **2010**, *46*, 4375–4377.
- (43) Pham, H. D.; Pham, V. H.; Cuong, T. V.; Nguyen-Phan, T.-D.; Chung, J. S.; Shin, E. W.; Kim, S. *Chem. Commun.* **2011**, *47*, 9672–9674.